PathFinder — Project Proposal

1. Introduction

This document serves as the formal proposal for the PathFinder project — a comprehensive platform designed to revolutionize how students discover and apply for academic and professional opportunities. The proposal outlines the motivation, scope, technical approach, and expected outcomes of the project.

2. Background

The modern academic landscape presents students with an overwhelming number of opportunities: internships at technology companies, scholarships from foundations, research grants from universities, and hackathons organized by developer communities. However, the infrastructure connecting students to these opportunities is fragmented and inefficient.

Students typically rely on:

- University career portals with limited listings
- Generic job boards that mix professional positions with student opportunities
- Social media channels where opportunities are shared inconsistently
- Word-of-mouth recommendations from peers and mentors

This fragmentation leads to missed deadlines, unsuitable applications, and a general sense of being overwhelmed. On the other side, organizations posting these opportunities receive hundreds of applications with no efficient way to assess candidate suitability.

3. Need for the System

Student Pain Points

1. **Information Overload**: No single source aggregates all opportunity types.
2. **Manual Tracking**: Students lose track of applications, deadlines, and statuses.
3. **Generic Applications**: Without intelligent matching, students apply broadly rather than strategically.
4. **Document Burden**: Writing unique SOPs and cover letters for each application is time-consuming.
5. **No Feedback**: Students rarely understand why they were accepted or rejected.

Organization Pain Points

1. **Application Volume**: Reviewing hundreds of applications manually is resource-intensive.
2. **Candidate Assessment**: No automated way to evaluate how well a candidate's profile fits the opportunity requirements.
3. **Communication Gap**: No built-in channel to communicate status changes to applicants.

4. Existing Systems and Their Limitations

Platform	Strengths	Limitations
LinkedIn	Large user base, professional networking	Not designed for student-specific opportunities; no matching engine
Internshala	India-focused internships	Limited to internships; no AI features; no scholarship/grant support
Scholly	Scholarship discovery	Only scholarships; no application tracking; limited AI
Handshake	University-integrated	Requires university partnerships; US-centric
DevPost	Hackathon listings	Only hackathons; no student profile management

PathFinder uniquely combines all opportunity types (Internships, Scholarships, Grants, Hackathons) into a single platform with AI-powered features that none of the existing systems offer comprehensively.

5. Proposed System

PathFinder is a full-stack web application built with the following high-level architecture:

- **Frontend**: React.js with TypeScript, styled with a custom CSS design system featuring glassmorphism and dark mode aesthetics.
- **Backend**: Node.js with Express.js, using Prisma ORM for type-safe database operations.
- **Database**: SQLite for portable, zero-configuration data storage.
- **Authentication**: JWT-based stateless authentication with bcrypt password hashing.
- **AI Integration**: External LLM API integration for intelligent features, with robust mock fallbacks.

The system supports three distinct user roles:

1. **Students**: Can build profiles, upload resumes, browse opportunities, receive AI-generated match scores, generate application documents (SOP, Cover Letter, Essay, Personal Statement), track application statuses, and submit support tickets.
2. **Organizations**: Can create company profiles, post opportunities, review applicants with match scores, download resumes, update application statuses (Accept, Reject, Shortlist), and view applicant summaries.
3. **Administrators**: Can view platform analytics, manage users (change roles, delete accounts), monitor system health, review audit logs, manage support tickets, and export data as CSV.

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6. Benefits

For Students

- **Time Savings**: AI-generated documents reduce application preparation time by an estimated 60%.
- **Better Targeting**: Match scores help students focus on opportunities where they have the highest likelihood of success.
- **Transparency**: Real-time application tracking with timeline events provides full visibility into the review process.
- **Profile Strength Analysis**: The profile completeness checker guides students to build stronger profiles.

For Organizations

- **Efficient Screening**: AI match scores and candidate summaries enable rapid assessment of applicant suitability.
- **Reduced Administrative Burden**: Batch status updates and automated notifications streamline the review process.
- **Quality Applicants**: The matching engine attracts more qualified applicants by ensuring better alignment.

For Administrators

- **Platform Oversight**: Real-time analytics dashboard with user metrics, application breakdowns, and system health monitoring.
- **Audit Trail**: Comprehensive audit logging for compliance and security review.
- **Issue Resolution**: Integrated ticketing system for direct communication with users.

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7. Objectives

1. Develop a production-ready web application with role-based access control.
2. Implement an AI-powered matching engine that scores student-opportunity compatibility.
3. Build AI-assisted document generation for SOPs, cover letters, essays, and personal statements.
4. Create a comprehensive admin dashboard with analytics, user management, and system monitoring.
5. Deploy the application to cloud infrastructure accessible via public URLs.
6. Produce complete developer documentation enabling future contributors to understand and extend the system.

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8. Technologies Overview

Layer	Technology	Purpose
Frontend Framework	React.js (via Vite)	Component-based UI development
Language	TypeScript	Static typing for reliability
Styling	Vanilla CSS	Custom design system with CSS variables
Icons	Lucide React	Consistent, modern iconography
Routing	React Router DOM	Client-side page navigation
State Management	React Context API	Global auth and theme state
Backend Framework	Express.js	RESTful API server
ORM	Prisma	Type-safe database queries
Database	SQLite	Lightweight relational database
Authentication	JWT + bcrypt	Secure token-based auth
File Handling	Multer	Resume and image upload processing
Email	Nodemailer	Password reset emails
AI Integration	External LLM API	Match scoring and document generation
PDF Processing	pdf-parse	Resume text extraction
Frontend Hosting	Vercel	Static site deployment
Backend Hosting	Render	Node.js application hosting

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9. Expected Outcome

Upon completion, PathFinder will be a fully deployed, publicly accessible platform where:

- Students can register, build comprehensive profiles, discover matched opportunities, generate AI-assisted application documents, and track their application journey.
- Organizations can post opportunities, receive and review applications with AI-generated match scores, and manage candidate pipelines.
- Administrators can monitor platform health, manage users, review analytics, and handle support requests.
- Future developers can clone the repository, follow the documentation, and extend the platform with new features.

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10. Future Scope

Version	Features
v2.0	Mobile application (React Native), AI interview preparation, calendar integration
v3.0	Government API integration, university portal integration, AI career roadmaps
v4.0	Multi-language support (i18n), real-time messaging, video interview scheduling

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End of Project Proposal